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ORIGINAL PAPER

The Changing Nature of Homicide in the Twin Cities: An Interrupted Time Series Analysis of the Police Murder of George Floyd

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Abstract

Objectives This study tests whether the 2020 police murder of George Floyd produced measurable changes in the quantity and qualities of homicide in Hennepin and Ramsey Counties, Minnesota—the epicenter of the ensuing social unrest—and whether these effects were moderated by neighborhood disadvantage or mediated by changes in police activity.

Methods We constructed a weekly panel of 472 census tracts from 2018–2022 (N= 123,192) using homicide data from the Minnesota Bureau of Criminal Apprehension. Interrupted time-series (ITS) models estimate post-event effects, controlling for COVID-19 caseloads, policy interventions, mobility, weather, and autoregressive structure. Random-coefficient ITS models assess spatial heterogeneity; a causal mediation sub-analysis of Minneapolis evaluates the contribution of de-policing to post-event homicide changes.

Results The homicide rate increased 2.5-fold—from 0.11 to 0.28 per 100,000 residents—immediately after Mr. Floyd's murder and remained elevated for 136 weeks, yielding an estimated 183 excess homicides. This effect persisted after adjustment for pandemic covariates. Argument-related killings showed the strongest post-event increase, followed by felony-crime homicides; domestic cases were unchanged. Adult perpetration and victimization rose significantly, while juvenile-involved homicides did not. The post-murder effect varied across space, with larger increases in tracts characterized by higher concentrated disadvantage. In Minneapolis, reduced police stops mediated 25% of the total effect.

Conclusions The police murder of George Floyd triggered a sustained escalation in homicide in the Twin Cities, driven by situational and interpersonal violence concentrated in disadvantaged neighborhoods. De-policing explained part of the increase. Findings highlight how high-profile police violence can generate collateral criminogenic effects and reinforce spatial inequality.

Keywords Homicide · Police violence · Interrupted time series · De-policing · Spatial analysis

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Introduction

The year 2020 brought overlapping institutional shocks to the United States. As the COVID-19 pandemic disrupted work, schooling, mobility, and social services, the police murder of George Floyd ignited a national *reckoning* over policing and criminal justice legitimacy (Phelps 2024). Homicide rates surged across the country, rising roughly 30 percent—the largest single-year increase in more than a century—before declining after 2022 (Gramlich 2021; Council on Crime and Justice 2023; Acharya & Morris 2024). Yet this national pattern masked sharp geographic and social inequalities: the recovery has been uneven, and the burden of lethal violence remains concentrated in historically disadvantaged neighborhoods (MacDonald et al. 2022; Ratcliffe & Taylor 2023; Light & Vachuska 2024).

Scholars have converged on several mechanisms to explain these shifts. Pandemic closures and mobility restrictions altered routine activities, reducing guardianship and reorganizing exposure patterns (Lopez & Rosenfeld 2021; Rosenfeld & Lopez 2020). Steep contractions in proactive enforcement may have weakened deterrence and the perceived certainty of sanctions (Nix et al. 2024; Lind et al. 2024). At the same time, the legitimacy crisis following Floyd's murder eroded trust in the criminal justice system, reducing public cooperation and undermining both formal and informal control (McCarthy et al. 2020). These institutional disruptions unfolded amid sharp increases in firearm purchasing and carrying, likely raising the lethality of everyday conflicts (Densley & Peterson 2024; Lopez & Rosenfeld 2021; Schleimer et al. 2021).

Beyond opportunity structures and institutional dynamics, pandemic conditions intensified psychological strain and weakened informal controls within households and neighborhoods. General strain and social control theories predict that acute stressors (e.g., job loss, financial precarity, social isolation, and family conflict) lower thresholds for aggression and impulsive violence (Agnew 1992; Hirschi 1969; Sampson & Groves 1989). National surveys in mid-2020 documented elevated anxiety, depression, trauma-related stress, and substance use—especially among young and less-educated adults—alongside increases in alcohol consumption (Czeisler et al. 2020; Pollard et al. 2020; Vahratian et al. 2021). Domestic violence rose across multiple jurisdictions during periods of lockdown (Leslie & Wilson 2020; Piquero et al. 2021). Taken together, these dynamics reflect a broader strain-and-stress context that may have amplified routine conflicts and eroded informal social control (Densley & Peterson 2024).

Against this national backdrop, the Minneapolis–Saint Paul metropolitan area offers a uniquely powerful setting for examining how these mechanisms interacted to shape lethal violence (Larson et al. 2023; Peterson & Densley 2024; Boehme et al.2022). Hennepin and Ramsey Counties experienced one of the nation's most visible institutional shocks: Floyd's murder triggered unprecedented local protests, rapid organizational changes within the Minneapolis Police Department, and marked declines in public cooperation and police proactivity (Phelps 2024). The region also exhibits pronounced spatial inequality and longstanding racialized disparities in housing, exposure to violence, and police contact (Larson et al. 2023)—conditions under which theories of routine activities, strain, spatial concentration, and legitimacy-based withdrawal make the clearest divergent predictions. In addition, unlike most U.S. cities, the Twin Cities maintain exceptionally high-resolution homicide data, and geocoded police-stop data that enable precise temporal and spatial modeling. The combination of high quality data and a strong-case setting are likely to provide upper-bound estimates of how large-scale institutional shocks can alter the quantity, composition, and spatial distribution of homicide, yielding insights that are likely to be attenuated in contexts where disruptions were less acute or data less granular.

Literature Review

Spatial Inequality, Routine Activities, and Pandemic Shocks

Foundational research identifies three durable structural determinants—poverty and educational disadvantage, racial composition, and family disruption—that anchor baseline homicide risk across places (Tcherni 2011). Macroeconomic conditions and market dynamics, such as economic downturns, further explain co-movement between acquisitive crime and homicide (Rosenfeld 2009). These macro-structural accounts clarify why the institutional shocks of 2020 represent not an anomaly, but a discontinuity imposed upon long-running trends in gun availability, substance markets, and social disorganization and collective efficacy (Council on Crime and Justice 2023; Tuttle 2025). At the same time, situational theories emphasize how such structural shocks interact with everyday routines and guardianship patterns.

Routine activity theory posits that crime occurs where motivated offenders and suitable targets converge in the absence of capable guardians (Cohen & Felson 1979). Pandemic-era closures and mobility suppression reshaped those convergences, increasing opportunities for situational violence, particularly in disadvantaged areas (Lopez & Rosenfeld 2021; Rosenfeld & Lopez 2020). Recent evidence from mobility and online-crime data confirms that COVID-19 restrictions systematically restructured daily routines and opportunity structures, supporting routine-activity explanations for pandemic-era crime variation (Johnson & Nikolovska 2024). The criminogenic effects of weakened formal and informal controls under acute stress are well documented (Hirschi 1969; Sampson & Groves 1989; Agnew 1992); social strain and despair may have increased volatility and self-directed or impulsive violence during this period (Peterson & Densley 2024). Pandemic-related school closures likely amplified these dynamics by removing structured supervision, peer regulation, and adult guardianship for adolescents (Oosterhoff et al. 2022). Routine activity perspectives therefore predict that violence involving young people—particularly unstructured, peer-conflict incidents—would increase during prolonged closures and mobility suppression. City-specific analyses show contemporaneous spikes in shootings consistent with these pressures (Boehme et al. 2022; Drake et al. 2022; Wolff et al. 2022).

Moreover, violent events are rarely isolated; they diffuse spatially across adjoining neighborhoods through social and retaliatory networks (e.g., Papachristos 2009), reinforcing the need for tract-level, time-resolved models (Cohen & Tita 1999). At finer spatial scales, a small fraction of places account for a disproportionate share of crime, a stable empirical regularity known as the “law of crime concentration” (Weisburd 2015). This concentration underscores that structural shocks are unlikely to be spatially uniform and motivates models capable of detecting heterogeneous post-event effects within and across neighborhoods.

Research suggests the 2020 homicide spike was steepest in neighborhoods already burdened by structural disadvantage and environmental disorder, compounding existing disparities in safety, health, and political power (MacDonald et al. 2022; Ratcliffe & Taylor

2023; White et al.2022). In Minneapolis, post-Floyd gun violence clustered spatially and temporally, with historically Black, disadvantaged spaces experiencing the brunt of the spike in gun assaults (Larson et al. 2023). These inequalities had measurable demographic consequences, including widening Black–White life-expectancy gaps among men (Light & Vachuska 2024) and community-level mental-health harms (Santaularia et al. 2025). Such stratification interacts with policing, as disadvantaged neighborhoods are both more heavily surveilled and more vulnerable when legitimacy collapses (Sampson & Bartusch 1998; Kirk & Papachristos 2011).

Policing and De-policing: Violence, Deterrence, and Legitimacy

The evidence that certainty—more so than severity—of punishment deters crime is robust (Nagin 2013). Policing can affect homicide through visibility (certainty), incapacitation, and targeted tactics. Randomized hot-spots experiments and systematic reviews demonstrate that focused patrol and problem-oriented strategies reduce crime in small geographic areas (Sherman & Weisburd 1995; Braga et al. 2014, 2019). Yet how police deploy their resources matters: Spelman (2017) shows that tactical quality and deployment decisions often outweigh simple headcount increases. Relatedly, optimizing patrol and district design raises both equity and legitimacy concerns that are operational as much as normative (Liberatore et al. 2022).

High-profile police killings can trigger dual reactions—legitimacy shocks among the public (legal cynicism and reduced cooperation) and behavioral pullbacks within departments (declines in proactive activity). In Minneapolis specifically, the police workforce contracted sharply following Mr. Floyd's murder. Between 2020 and 2022, 273 officers left the department—about 40% of its sworn strength—through retirements, resignations, and a surge in workers' compensation claims (Kolls 2023). Evidence from the Ferguson era shows that public scrutiny following police violence produced measurable reductions in officer-initiated stops nationwide, accompanied by modest short-term increases in violent and property crime (Powell 2023). Similarly, Vandegrift & Connor (2020) find that a police use-of-force incident predicted nearly a two percent rise in murders two months later, with effects stronger pre-Ferguson but reversing in some later contexts. Yet other multi-city analyses find mixed or null average effects (Pyrooz et al. 2016; Rosenfeld & Wallman 2019).

At finer scales, neighborhood-level studies report that sharp drops in stops correlate with increases in violent and property crime (Nix et al. 2024) and that Minneapolis experienced contemporaneous jumps in carjacking and homicide after George Floyd's murder (Lind et al. 2024). Countervailing evidence complicates these patterns: Moyer (2022) finds no decline in community reliance on police services after a death-in-custody event, and Simpson & Kirk (2023) report no contagion of officer misconduct across networks, suggesting that organizational responses vary substantially.

Taken together, the literature supports the situational deterrence value of proactive policing but underscores its heterogeneity. The balance of evidence suggests that policing pullbacks do occur after legitimacy crises, but their crime consequences are contingent on local context and several other mechanisms this study investigates at the neighborhood level.

The Current Study

Several gaps in the literature remain. First, prior work overwhelmingly emphasized homicide *quantities,* leaving the *qualities* of lethal events, such as circumstance (e.g., argument vs. felony), victim–offender age profiles, and weapon use, largely unexamined at scale. These dimensions are central to theorizing how routine activities, strain, and cultural conflict responded to 2020's disruptions. Second, existing city-level aggregation masks the well-documented spatial concentration of crime (Weisburd 2015), tract-level heterogeneity and spatial diffusion dynamics central to concentrated harm (Cohen & Tita 1999). This obscures whether post-2020 changes were uniform or disproportionately borne by structurally disadvantaged neighborhoods, and the current study offers a spatially resolute test of effect heterogeneity across space and community context. Third, although researchers have interpreted 2020 trends through frameworks of depolicing and legitimacy shocks (Powell 2023; Roman et al. 2025), few studies have tested whether measurable changes in officer activity statistically mediated shifts in lethal violence. By replicating prior work in a new city (e.g., Nix et al. 2024) but integrating high-resolution homicide circumstances, neighborhood-week measures of proactive activity within interrupted time-series and multilevel frameworks that statistically adjust for pandemic covariates (e.g., COVID-19 cases, policy timing, mobility) and seasonality, our study links three fragmented literatures and offers a more complete account of how institutional shocks altered both the quantity and nature of homicide.

Here, we analyze hand-coded homicide data for Hennepin and Ramsey Counties (2018–2022) to address three questions: (1) Did homicide levels and trajectories change following the police murder of George Floyd? (2) Did effects vary across neighborhoods, particularly by concentrated disadvantage? And (3) To what extent do within-neighborhood changes in proactive policing (stops) mediate any post-event effects in Minneapolis? We estimate tract-week interrupted time-series and random-coefficient panel models that (a) identify level and slope changes at the Floyd event, (b) allow effects to vary by concentrated disadvantage, (c) incorporate spatial exposure and diffusion, and (d) explicitly test qualitative shifts in homicide.

To the latter point, each mechanism implies different observable shifts. If routine-activity disruption from school closures and mobility loss drove the spike, juvenile-involved or opportunistic street conflicts should have risen. If pandemic-induced strain and family stress dominated, domestic homicides would likely have increased. If economic deprivation was central, felony- or robbery-related killings should have surged. By explicitly modeling the qualitative dimensions of crime—circumstance, victim–offender, age-role interaction, and weapon use—we test which mechanisms are most consistent with observed post-event changes.

Substantively, we provide one of the most granular tests to date of how intertwined routine-activity shocks, policing pullback, and legitimacy map onto both the quantity *and* quality of lethal violence in a city at the center of the 2020 crisis. Theoretically, we bridge structural baselines (Tcherni 2011; Rosenfeld 2009), situational opportunity (Cohen & Felson 1979), deterrence and tactics (Nagin 2013; Sherman & Weisburd 1995; Braga et al. 2019; Spelman 2017), spatial diffusion (Cohen & Tita 1999), and the legitimacy/legal-cynicism nexus (Sampson & Bartusch 1998; Kirk & Papachristos 2011). Methodologically, we integrate high-resolution place–time modeling with mediation diagnostics to parse mechanisms rather than attributing the 2020 spike to any single cause (Nix et al. 2024; Lind et al. 2024; Moyer 2022; Vandegrift & Connor 2020; Simpson & Kirk 2023; Council on Crime and Justice 2023; Acharya & Morris 2024).

Data and Methods

Panel Data

Hand-Coded Bureau of Criminal Apprehension (BCA) Homicide Data

Our focal dependent measures are constructed from a unique, hand-coded Minnesota Bureau of Criminal Apprehension (BCA) homicide dataset. The BCA provided the population of homicides recorded in Hennepin and Ramsey Counties between 2018 and 2022 based on National Incident-Based Reporting System (NIBRS) records. To enhance contextual accuracy, additional information was gathered from media reports, police records, and trial documents. A total of 993 homicide cases were coded across 75 variables, capturing information on timing and location, victim and perpetrator demographics, circumstances of the homicide, and trial outcomes.

Data coding was conducted by trained, advanced undergraduate research assistants with prior experience in data classification. Coders completed rigorous training, including coding ten cases to establish reliability. Each case was independently coded by two coders to minimize bias, with discrepancies reviewed and resolved by the principal investigators. A lead research assistant (a former Minneapolis police officer), conducted a final quality control review, and the research team met weekly to address coding challenges, ensuring methodological consistency. This coding process aligns with established best practices, including those used in The Violence Project Database of Mass Shooters (Peterson & Densley 2025).

Using the geocoded XY coordinates of each coded homicide, we spatially located each homicide within Hennepin and Ramsey County Census Tracts via a spatial join with an intersection predicate. We then aggregated homicides (stratifying by various homicide characteristics) within each *tract-week* to create a tract-week panel of homicide counts consisting of 472 Census Tracts (identified with a spatial filter with a within predicate using a dissolved Hennepin/Ramsey county shapefile) and 261 ISO weeks for 123,192 observations within Hennepin and Ramsey counties in the full panel.

Interrupted Time Series Indicators

In our interrupted time-series design (discussed further below), the key time indicators are a baseline weekly linear trend (T), an exposure indicator of the police murder of Mr. Floyd in the week of 5/25/2020 (Post-Murder), and the difference in the linear time trend post-murder (T Post-Murder). The weekly time trend captures the overall linear trend in homicides across 2016–2020, the binary exposure variable indicates the discontinuous change in the week of the police murder of Mr. Floyd, and the post-killing term describes changes in the linear trend in the homicide rate after the murder.

COVID-19 Pandemic, Movement, and Related Policy

In addition to our focal time measures that capture the time structure of homicide changes before and after the murder of Mr. Floyd, we adjust for time-varying changes in COVID-19-movement, caseload, and related policy to further improve our identification of the post-murder effect in our interrupted time series design. We measure weekly COVID-19 case counts and mortalities for Hennepin and Ramsey counties using the New York Times' Coronavirus Data in the United States database (New York Times 2025). Specifically, we filter the county-level database to include only Hennepin and Ramsey counties, and aggregate daily case counts for each week (weeks before COVID-19 onset were set to zero).

We adjusted for COVID-19 time-varying movement changes using Google's COVID-19 Community Mobility Reports (CMR). Google's CMR collates data from those accessing Google applications (i.e., Google Maps) with smartphones who allow recording of 'location history' (Google 2023). Individual data are aggregated to the sub county-level (i.e., city), and relative percentage change from median baseline mobility on 1/3/2020–1/6/2020 are categorized into six categories: 'retail and recreation', 'parks', 'groceries and pharmacies', 'workplaces', transport 'transit' hubs and 'residential' areas. We filter these data to include the Hennepin and Ramsey County subregions, and aggregate weekly mean mobility across the six categories. Weeks before 2020 and after 10/15/2022 are not available in these data, and we set these weeks to zero (i.e., baseline mobility). Therefore, this time-varying control captures movement changes during the focal COVID-19 pandemic period, as well as during the period of local unrest following the police murder of Mr. Floyd.

In addition to COVID-19 caseloads and movement patterns, we create time-varying indicators of COVID-19 state policy to further adjust for pandemic-related influences on homicide. Specifically, we use 5 binary "on/off switches" for the following state-level policy changes:

Stay at Home Order: 3/28/2020–5/28/2020
State of Emergency Order: 3/13/2020–7/1/2021
General Public Access to COVID-19 Vaccine: 3/30/2021 onwards
School Closure: 3/18/2020–2/22/2021
Business Closure (restaurants, bars, movie theaters, gyms): 3/27/2020–6/10/2020; 11/21/2020–1/11/2021

These time-varying policy indicators adjust for changes in homicide related to significant policy events during the COVID-19 pandemic and related patterns of social interaction not captured by our average mobility measure discussed above. These data vary within but not between tracts.

Department of Natural Resources Weather Data

Previous scholarship indicates that homicides exhibit a seasonal pattern (e.g., McDowall & Curtis 2015), and we merge measures of weather seasonality onto the panel. These include the weekly maximum temperature (degrees Fahrenheit), snowfall (in.), and precipitation (in.) from the Minnesota Department of Natural Resources as measured at the Minneapolis/

St. Paul Threaded Record station (Minnesota DNR 2025). These data also vary within but not between tracts.

American Community Survey 5-Year Estimates

To assess the spatial heterogeneity of the post-murder effect, we access the Census Bureau's API via the `tidycensus` package in R (Walker 2025) to download yearly American Community Survey 5-year estimates by tract-year. Specifically, we include measures of the unemployment rate, poverty rate, female headed household rate, and no high school diploma rate which comprise our measure of concentrated disadvantage as described below. These data vary both within (yearly) and between tracts.

Local Police Stop Data

On the Minneapolis subpanel (i.e., a spatially filtered dataset of tract-weeks within the city boundaries of Minneapolis), we merged on weekly counts of police stops as reported by the Minneapolis Police Department. We located each reported stop within each tract-week, and merge onto each tract-week in the Minneapolis subpanel. We followed the measurement strategy of Nix et al. (2024) to create a rolling deviation measure of police stops by calculating the extent to which police stops deviated from the tract-level status quo. We did this by generating 3-week rolling average deviations in observed police stops from the same 3-week rolling average in 2018–2019 weighted by recency. This measure has many advantages over a standard time-varying police rate, including 1) tempered weekly fluctuations in police stops from the use of a moving average, 2) recency-weighted baseline observations to account for long-term temporality, 3) a comparison to a baseline level of police stops that is consistent with the routine activities and rhythms of the year, and 4) a counterfactual approach that removes the simultaneity between policing and homicide, as the prior year's police stops cannot be determined by homicide in the focal year.

Analytical Strategy

Descriptive

We first create descriptive time series plots using an aggregated time series that aggregates our measures over tracts. Time-series plots of overall homicide incidence per 100,000 residents from 2018–2022 capture temporal fluctuations in the homicide rate in the quantity of homicides, displayed in Fig. 1. We then create time-series plots of homicide rates per 100,000 stratifying by qualities of the homicides across circumstance (argument, crime, domestic), the interaction of age (juvenile, adult) and role (victim, perpetrator), and weapon (firearm, other) faceted in Fig. 2. Discussion of the magnitude of pre/post mean differences is augmented with Welch's T-Tests of the homicide rate before and after the event assuming unequal variances using the Satterthwaite approximation for degrees of freedom (Satterwaite 1946) (Figs. 3 and 4). In our analyses of the full unaggregated panel, we describe the spatiotemporal variation in homicide rates in Appendix Fig. 6 by constructing faceted choropleth maps, using Census Bureau TIGER/Line shapefiles (Census Bureau 2025), that

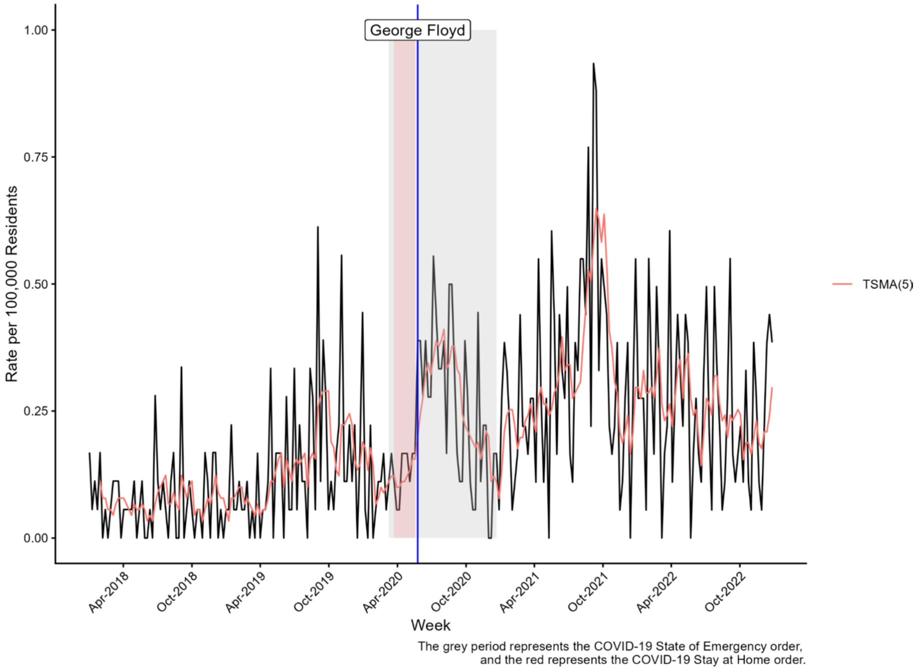

Fig. 1 Violence Prevention Project Homicide Data

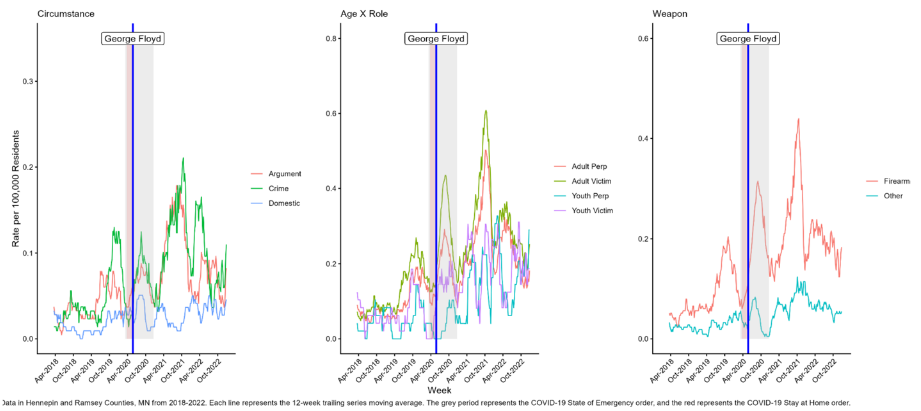

Fig. 2 TSMA(12) of weekly homicides by circumstance, age X role, and weapon

depict the between-tract variation in three temporal windows: Pre-Murder, 0–3 Months Post-Murder, and 3+Months Post-Murder. In our depolicing sub-analysis the temporal variation in the police deviation measure is visualized with a time series plot (Fig. 5) of deviations relative to the weighted baseline period. Each grey line represents a tract in Minneapolis, and the red solid line depicts the median TSMA(3) deviations in police stops, and the dotted series represent the 90th percentile and 10th percentile of deviations, respectively.

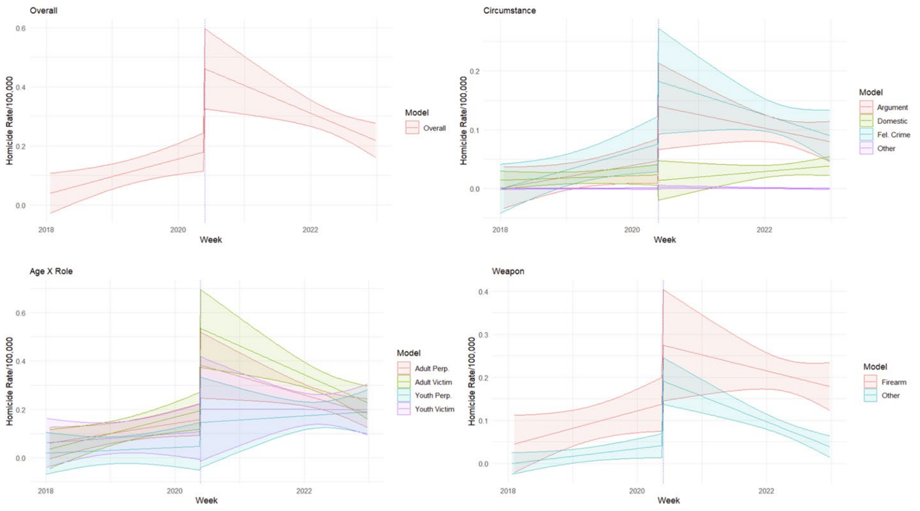

Fig. 3 Marginal ITS effect plots of homicide by specification

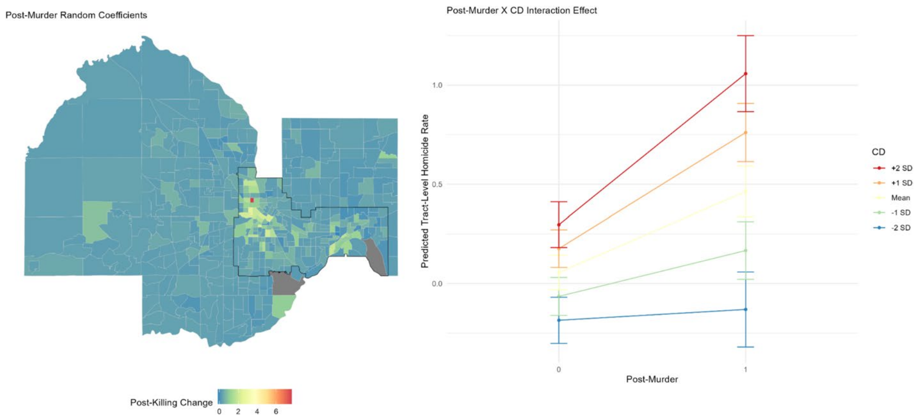

Fig. 4 Spatial heterogeneity of post-murder effect

Explanatory

We then estimate autoregressive interrupted time series (ITS) models on the aggregated week-level time series data across Hennepin and Ramsey counties. ITS designs compare the levels of outcomes after a treatment or intervention to the outcome level in the pre-intervention period. The pretreatment trend is assumed to serve as the counterfactual should treatment not have occurred (i.e., the trend in homicide rate in the

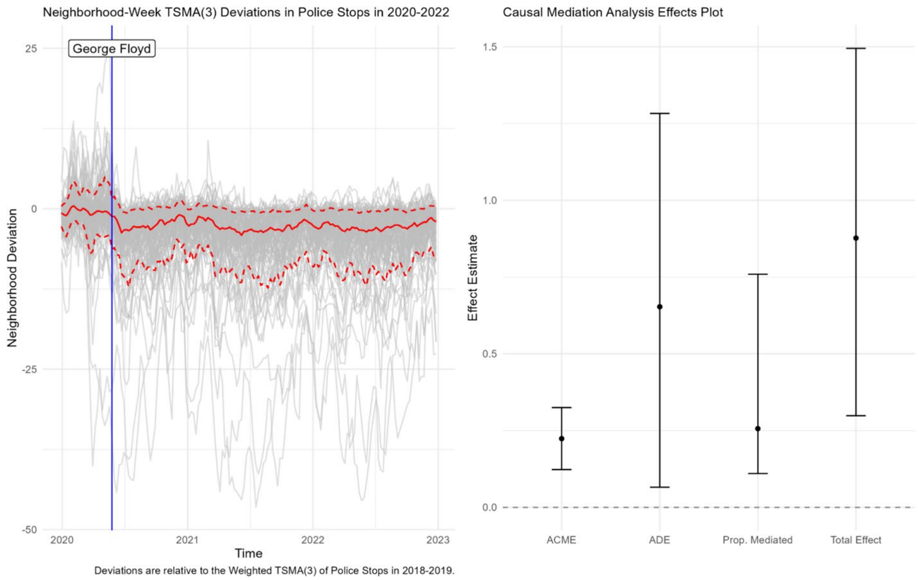

Fig. 5 Minneapolis police stop mediation analysis

counterfactual scenario in which the police murder of Mr. Floyd did not occur). The design exclusively uses within-unit over time variation, so time-stable confounders are uncorrelated with the time-varying treatment. However, our design is susceptible to time-varying unobserved heterogeneity, which can vary alongside treatment timing, and we therefore include a suite of time-varying controls to strengthen our identification strategy in the ITS design. Further, the design is susceptible to temporal autocorrelation in the series, and therefore we include temporal autoregressive lags of the homicide rate in the *t-lag* week to adjust for the impact of homicide levels on contemporaneous homicides, effectively controlling for the possibility of the focal event timing being confounded with recent changes in homicide. We choose the number of lags based on partial autocorrelation function tests of autocorrelation in model residuals. In sum, our causal identification approach assumes that the police murder of Mr. Floyd, and its consequences, are the only exposures that change at the time of the event, net of the observed time-varying covariates.

The ITS design is also susceptible to bias if the series is nonstationary (i.e., does not have constant mean and variance). We include a deterministic trend (t) that captures the average weekly change across the series, but this term does not capture temporal fluctuations that are stochastic (i.e., random shocks that accumulate and permanently alter levels and variation of the series), as opposed to deterministic (i.e., random shocks that are temporary and not baked into levels or variation permanently). This can lead to “spurious regression” wherein the timing of the intervention (or time variation in the other covariates) can correlate with temporal drift caused by random shocks in homicide, even though intervention and outcome may not be causally related (Enders 2015). Therefore, we perform Augmented Dickey-

Fuller (ADF)[1] and Kwiatkowski-Phillip-Schmidt-Shin (KPSS)[2] tests to assess residual stationarity after inclusion of the deterministic trend and covariates. Across all pooled and panel specifications, both tests confirm series stationarity around the deterministic trend for all models (see diagnostics in Tables 1, 2, 3 and 4). Accordingly, all models are estimated in levels as opposed to changes through differencing.

Importantly, we attempt to extricate the effect of the COVID-19 pandemic's progression from the police murder of Mr. Floyd by including multiple crime-relevant, time-varying features of the pandemic progression as described in the measures above (i.e., cases, mortality, mobility, policy). While the COVID-19 measures exhibit changes during the week of the murder,[3] we only extricate the effect of Mr. Floyd's murder from the movement of the pandemic to the extent that our measures capture covariation between the pandemic measures and the homicide rate. Further, our time-varying measures of the pandemic vary at the week-level but are constant across tracts, which leaves our identification susceptible to between-tract heterogeneity in pandemic experiences in our panel models. Our pooled models can generally be expressed as follows:

$$y_t = \beta_0 + \beta_1 Time + \theta Event_t + \beta_2 TimePost_t + \phi X_t + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \rho_3 y_{t-3} + \epsilon_t \quad (1)$$

where θ represents the focal parameter of interest: the change in the time series of homicide rates in response to the police murder of Mr. Floyd. We specify models for the overall homicide rate (Appendix Table 1) and also for each subcategory in the coded homicide variables of circumstance, the interaction of age and role, and weapon (Appendix Tables 2, 3 and 4). In the main manuscript, we present these models by constructing marginal effects from each ITS model, which plot the predicted homicide rates at each time point across the series in accordance with the estimated ITS coefficients, with covariates held at their mean or referent values. The ribbons on each ITS model indicate 95% confidence intervals. All time series models are estimated via OLS with the `lm` function in R.

To assess the spatial heterogeneity in the post-murder effect, we estimate random coefficient interrupted time series panel models using the fully constructed tract-week panel with random census tract intercepts and census tract random Event (i.e., post-murder) coefficients. This ITS specification using the panel data allows the intercept and focal Event coefficients to vary by tract allowing us to a) account for baseline tract differences in homicide rates (random intercept), as well as b) examine the spatial heterogeneity in the post-killing effect (random coefficient) across tracts. The main RE panel model is specified as follows:

$$\begin{aligned} y_{ti} &= \beta_{0i} + \beta_1 Time + \theta_i Event_t + \beta_2 TimePost_t + \phi X_{ti} + \rho y_{t-1} + \rho y_{t-2} + \rho y_{t-3} + \epsilon_{ti} \\ \beta_{0i} &= \gamma_{00} + u_{0i} \\ \theta_i &= \gamma_{10} + u_i \end{aligned} \quad (2)$$

[1] The ADF tests evaluate whether a time series contains a unit root by testing whether the lagged level term significantly predicts changes in the series after accounting for the deterministic components and short-run autocorrelation dynamics. The null hypothesis is series non-stationarity.

[2] KPSS tests evaluate whether a time series is stationary around a deterministic trend by examining whether the cumulative sum of residuals from a regression on deterministic components diverges more than expected under a stationary process. The null hypothesis is that the series is stationary.

[3] Specifically, the covid time-varying covariates with weekly resolution in this window (Year 2020 ISO Week 21—Year 2020 ISO Week 22) exhibit -15%, +8%, and -233% (-5.5% percentage point change from median baseline mobility) weekly changes respectively for COVID cases, COVID deaths, and average mobility respectively.

where θ_i represents the estimated post-killing increase in each Tract respectively.[4] Further, we specify the random effects models to have correlated random effects, which allow that the random intercepts and slopes may be related in some manner. These RE panel specifications include the full suite of controls for time-varying weather seasonality, and COVID-19 policy, case load, mobility, as well as three lagged AR terms to control for autocorrelation structure in the data. Full model estimates can be found in Appendix Table 6 and we visually present these focal random post-murder coefficients in the first panel of Fig. 5. This choropleth map visualizes the estimated post-killing change in each Tract (i.e., the estimated random coefficient—θ_i) across Hennepin and Ramsey County Census Tracts.

We also estimate a random coefficient model with a cross-level interaction between the post-murder indicator and an index of concentrated disadvantage to examine the spatial heterogeneity in the post-killing effect across communities, and to assess the moderating influence of structural racism and disadvantage on the effect of the police murder of Mr. Floyd (Appendix Table 6 Model 2). In other words, we explicitly test the extent to which the post-murder increase was different in areas of higher concentrated disadvantage and to what extent concentrated disadvantage may explain any post-murder effect heterogeneity. In the main manuscript, we construct an interaction plot within Fig. 4 depicting the model-predicted tract-level homicide rate before and after the police murder of George Floyd at varying levels of concentrated disadvantage (−2 SD, −1 SD, Mean, +1 SD, +2 SD). All random coefficient models are estimated via maximum likelihood using the `lme4` package in R.

Depolicing Sub-Analysis

To assess the extent by which depolicing may have operated as a mechanism by which the police murder of George Floyd impacted homicide rates, we conduct a sub-analysis using a subpanel consisting of tracts that are spatially within the city boundaries of Minneapolis. Due to the lack of systematic publicly available information from law enforcement agencies about police stops across the entire counties, we limit this subanalysis to only those tracts for which we have spatio-temporal records of police stops for this time period. Specifically, we conduct a causal mediation analysis of the extent to which the post-murder effect on homicide was mediated by changes in police stop deviations. Rooted in the counterfactual framework, the causal mediation analysis method (Imai et al. 2010) uses estimated potential outcomes from mediation and outcome models (Appendix Table 7) to estimate causal quantities of interest. Specifically, we use estimated potential outcomes from these models to estimate the total effect (TE), which is expressed as

$$TE = \frac{1}{N}\sum\nolimits_{i=1}^{N}\left(\widehat{Y}_i\left(T=1\right) - \widehat{Y}_i\left(T=0\right)\right) \tag{3}$$

The TE represents the average effect of the police murder of Mr. Floyd on the homicide rate. The average direct effect (ADE) is expressed as

$$ADE = \frac{1}{N}\sum\nolimits_{i=1}^{N}\left(\widehat{Y}_i\left(T=1, M=\widehat{M}_i\right) - \widehat{Y}_i\left(T=0, M=\widehat{M}_i\right)\right) \tag{4}$$

[4] This is effectively the estimated fixed effect (i.e., average event effect across tracts) added to the estimated random effect (i.e., the random deviation of the Tract slope from the average effect) for each Census Tract (*ui*) – sometimes referred to the unit-specific coefficients.

The ADE represents the average direct effect of the police murder of Mr. Floyd holding the mediator fixed (i.e., the effect of the murder that is not mediated by stops). The average causal mediation effect (ACME) is formulated as

$$ACME = \frac{1}{N}\sum\nolimits_{i=1}^{N}\left(\widehat{Y}_i\left(T = t, M = \widehat{M}_i\left(1\right)\right) - \widehat{Y}_i\left(T = t, M = \widehat{M}_i\left(0\right)\right)\right) \quad (5)$$

The ACME represents the difference in potential outcomes due to changes in the predicted mediation outcomes under the treatment and control conditions. This mediation analysis rests upon the sequential ignorability assumption (i.e., treatment is ignorable given covariates, and mediator is ignorable given treatment and covariates) and the linearity assumption. Standard errors of these causal quantities are estimated via quasi-Bayesian Monte Carlo simulation via the `mediation` package in R (Tingley et al. 2014). We present our mediation analysis in Fig. 5 through a coefficient plot depicting the estimated causal quantities discussed above alongside estimated standard errors.

Results

Temporal Pattern of Homicides

Changes in Homicide Quantity

The weekly homicide rate in Hennepin and Ramsey Counties increased sharply following the police murder of George Floyd, rising from an average of 0.11 to 0.28 per 100,000 residents—an increase of more than 2.5 times ($t(228.04) = 9.45$, $p < 0.05$). During the week of the murder, the rate climbed from 0.17 to 0.38 per 100,000 residents. By contrast, neither the March 2020 COVID-19 Stay-at-Home order nor the State of Emergency declaration from the Governor was associated with meaningful changes in homicide rate (Fig. 1).

Changes in Homicide Quality

Temporal trends stratified by homicide characteristics (Fig. 2) show that argument-related ($\mu_d = 0.05$, $t(234.64) = 4.52$, $p < 0.05$), felony-crime-related ($\mu_d = 0.06$, $t(223.34) = 4.69$, $p < 0.05$), and domestic ($\mu_d = 0.01$, $t(258.68) = 2.77$, $p < 0.05$) homicides all increased significantly after the event. Adult homicide perpetration ($\mu_2 = 0.15$, $t(218.84) = 7.79$, $p < 0.05$) and victimization ($\mu_d = 0.19$, $t(220.95) = 8.69$, $p < 0.05$) also rose markedly. Juvenile perpetration ($\mu_d = 0.10$, $t(209.23) = 3.66$, $p < 0.05$) and victimization ($\mu_d = 0.11$, $t(204.18) = 3.75$, $p < 0.05$) increased to a lesser extent. These descriptive differences suggest that both adult and juvenile-involved homicides increased in frequency, with larger absolute changes among adults.

Interrupted Time Series Models

Figure 3 presents the marginal effects from interrupted time-series (ITS) models for overall and subgroup homicide rates from 2018–2022. Models adjust for weather, sunset variation, COVID-19 cases and mortality, policy interventions, population mobility, and autoregressive error structure (see Appendix Table 1 for full specification).

In the overall model, the pre-treatment slope is positive and statistically significant ($\beta_1 = 0.001$, 95% CI [0.0003, 0.002]), indicating a modest upward trend in homicide before the police murder of George Floyd. The immediate post-murder level change (θ) was statistically significant and large ($\theta = 0.28$, 95% CI [0.132, 0.430]). Using 2020 ACS population estimates for Hennepin and Ramsey Counties, this corresponds to approximately five additional homicides per week (95% CI [2.38, 7.78]) relative to the counterfactual of a continued pre-murder trend (1,801,894/100,000*0.28). The change in slope after the event ($\beta_3 = -0.003$, 95% CI [–0.005, –0.001]) indicates a gradual decline following the spike, though rates did not revert to pre-murder levels within the observation window. Summing weekly effects yields approximately *183 excess homicides—about 26% of all post-murder incidents—beyond the projected baseline over the 136 weeks following the event* $\left(\sum_{t=1}^{136}\theta + \beta_3 * t\right)$.

We estimated subgroup marginal ITS models to investigate the *qualities* of homicides that drove this increase. These reveal heterogeneity by homicide circumstance (i.e., argument, domestic, felony crime, other), age-role interaction (adult, juvenile, victim, perpetrator), and weapon type (firearm, other). Among circumstances, argument-related homicides showed a significant increase ($\theta = 0.093$, 95% CI [0.011, 0.175]), while felony-crime-related homicides (e.g., drive by, robbery, gang-related) exhibited a similar but nonsignificant pattern ($\theta = 0.095$, 95% CI [–0.009, 0.200]).[5] Domestic ($\theta = -0.009$, 95% CI [–0.047, 0.028]) and other ($\theta = 0.002$, 95% CI [–0.00002, 0.00002]) homicides (e.g., hit-and-run, negligence) remained stable.

For age and role, both adult (18+) perpetration ($\theta = 0.225$, 95% CI [0.073, 0.376]) and adult victimization ($\theta = 0.340$, 95% CI [0.165, 0.516]) increased positively and significantly. Perhaps surprisingly, juvenile perpetration ($\theta = 0.0002$, 95% CI [–0.001, 0.001]) and juvenile victimization ($\theta = 0.0004$, 95% CI [–0.001, 0.002]) did not increase significantly. Weapon-stratified models show positive but uneven changes: firearm-related homicides rose modestly ($\theta = 0.137$, 95% CI [–0.006, 0.279]), and non-firearm homicides increased more substantially ($\theta = 0.150$, 95% CI [0.090, 0.211]), but neither reach statistical significance.

Collectively, these results indicate that the qualitative distribution of homicides changed after the police murder of George Floyd, with argument- and felony-crime-related incidents among adults driving the overall increase. Moreover, while both firearm and non-firearm homicides increased, the relative rise was larger for non-firearm incidents.

Spatial Heterogeneity in Post-Murder Effects

We assessed spatiotemporal variation by mapping tract-level homicide rates (Appendix Fig. 1) and estimating random-coefficient ITS models on a census tract–week panel (Appendix Table 6). This approach accounts for the nested data structure (i.e., weeks within tracts) and allows post-murder coefficients to vary across space. Figure 4 plots the tract-specific random effects, showing concentrated post-event increases in North and South Minneapolis (near what is now known as George Floyd Square) and East St. Paul, further validating

[5] Given we have a population of tract-weeks within Hennepin and Ramsey counties, we are less concerned with tests of statistical significance and focus more closely on effect magnitudes.

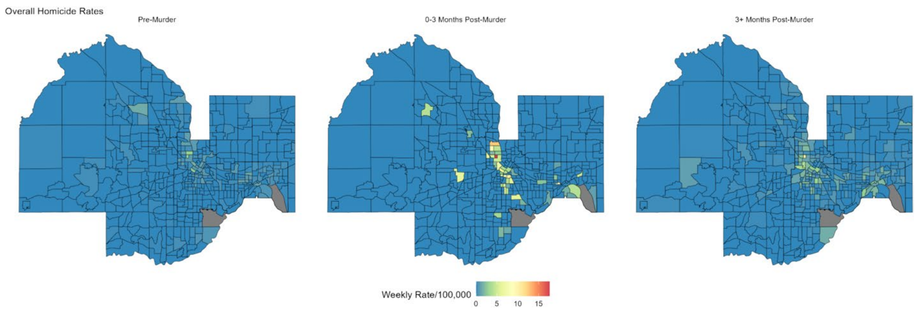

Fig. 6 Homicide rates by census tract and period, Hennepin & Ramsey counties 2018–2022

the patterns seen in Appendix Fig. 6. Interaction models (Fig. 4) between the post-murder term and concentrated disadvantage (CD) reveal significant moderation (β = 0.177, 95% CI [0.103, 0.248]). Predicted values indicate that homicide rates changed little in tracts at –1 to –2 SDs below the mean CD, but increased steeply at + 1 and + 2 SDs. These findings demonstrate that the post-murder homicide surge was spatially uneven and disproportionately concentrated in structurally disadvantaged neighborhoods.

Assessment of Depolicing Hypothesis

We next examined whether de-policing mediated the observed increase in homicide using a subpanel of Minneapolis tracts (2018–2022). Because systematic stop data were unavailable across both counties, police behavior was measured as the within-tract weighted moving-average deviation in weekly stops during the pandemic period (2020–2022) relative to 2018–2019. Figure 5 shows (a) neighborhood-specific weighted moving-average deviations in police stops, alongside the 10th, 50th, and 90th percentiles of the tract distribution, and (b) estimates from a causal mediation analysis treating the post-murder indicator as the exposure and deviations in police stops as the mediator. ITS models treating police stops as the outcome show an immediate decline of 4.4 stops per tract-week (95% CI [–4.6, –4.2]) after the murder, followed by a slow return toward baseline (β_3 = 0.012, 95% CI [0.009, 0.015]).[6] The mediation analysis estimates an average causal mediation effect (ACME) of 0.22 homicides per 100,000 (95% CI [0.13, 0.32]),[7] representing approximately 25% of the total effect (ATE = 0.87, 95% CI [0.30, 1.48]). Thus, reduced proactive policing explains a meaningful portion—but not the majority—of the post-murder increase in homicide, suggesting that broader social and institutional mechanisms were also at work.

[6] See Table 7 in the Appendix for full model specifications within the mediation analysis.

[7] The ACME is interpreted as the difference in potential outcomes due to changes in the predicted mediation outcomes under the counterfactual treatment and control conditions, holding fixed the treatment. In other words, the effect of the mediator that is induced via treatment assignment.

Conclusion

The United States experienced a historic spike in homicide during 2020 amid the overlapping crises of the COVID-19 pandemic and the social unrest following the police murder of George Floyd. Minneapolis–St. Paul, Minnesota was both the site of the murder and the epicenter of this unrest (Phelps 2024; Densley & Peterson 2024). Nationally, homicide rose roughly 30 percent—the largest one-year increase in more than a century—and remained elevated through 2022 before declining unevenly after that (Gramlich 2021; Council on Crime and Justice 2023; Acharya & Morris 2024). Against this backdrop, we examined how both the quantity and qualities of homicide changed in Minnesota's Hennepin and Ramsey Counties, isolating the effects of the police murder of George Floyd from concurrent pandemic-related factors.

Our analyses yield four central findings. First, interrupted time-series models show an immediate 2.5-fold rise in the weekly homicide rate following the murder and a sustained elevation over the subsequent 136 weeks. We estimate approximately 183 excess homicides are attributable to the murder and subsequent unrest, representing about one-quarter of all post-murder homicides. Neither the March 2020 stay-at-home order nor the state emergency declaration produced comparable changes, and the effect persisted after adjusting for COVID-19 caseloads, mortality, mobility, and weather. Second, the qualitative composition of homicide shifted: argument-related killings showed the strongest and statistically significant increase, with similar but nonsignificant patterns for felony-crime homicides, while domestic and "other" homicides remained relatively stable. Third, the post-murder period also saw large, significant increases in adult perpetration and victimization, with little change in juvenile-involved homicides and increases in both firearm and non-firearm killings. Fourth, these changes were spatially concentrated: tracts with greater concentrated disadvantage experienced the largest post-murder increases, clustering in North and South Minneapolis and East St. Paul. In Minneapolis, police stops fell sharply after the murder, and mediation models attribute roughly 25 percent of the total post-murder homicide effect to reduced proactive activity.

Taken together, these results suggest that the 2020–2022 rise in homicide in the Twin Cities was not a uniform breakdown of order, but a compositional and spatially uneven shift concentrated in everyday interpersonal conflicts. The prominent increase in argument-related homicides, alongside relatively stable domestic killings and youth-involved cases, points to a surge in situational disputes—public or semi-public encounters between adults that escalated more quickly and more lethally than before. This pattern is consistent with a context in which pandemic-related strain, disrupted routines, and heightened firearm availability lowered thresholds for aggression and made minor conflicts more likely to turn deadly, rather than a wholesale transformation of family violence or youth offending (Leslie & Wilson 2020; Rosenfeld & Lopez 2020; Lopez & Rosenfeld 2021; Piquero et al. 2021; Schleimer et al. 2021; Baumer & Staff 2025). This is in contrast to expectations of rising domestic violence during pandemic lockdowns.

The strong spatial concentration of the post-murder increase in disadvantaged neighborhoods underscores the need to interpret the homicide spike through the lens of cumulative inequality (Henderson et al. 2024). Long-standing racialized disparities in housing, exposure to violence, and police contact in Minneapolis–St. Paul meant that structurally marginalized communities entered 2020 with fewer buffers against institutional shocks

(Wrigley-Field et al. 2020). The Floyd murder and its aftermath did not create new zones of vulnerability so much as intensify existing ones. In this sense, our findings support accounts that conceptualize the homicide spike as a stress test of urban inequality: when institutional legitimacy erodes and routine supports fracture, the costs of violence are most visible where disadvantage, legal cynicism, and weak collective efficacy already converge (Sampson & Groves 1989; Tcherni 2011; Ratcliffe & Taylor 2023; Santaularia et al. 2025; Light & Vachuska 2024).

Our mediation analysis provides a more nuanced view of depolicing in this environment. The evidence that reduced proactive policing explains roughly one-quarter of the post-murder homicide effect confirms that organizational withdrawal and diminished guardianship were criminogenically meaningful, consistent with prior work on stop declines and crime increases during periods of heightened scrutiny (Nix et al. 2024; Powell 2023; Vandegrift & Connor 2020). At the same time, the fact that three-quarters of the effect remains unexplained by stops alone cautions against accounts that attribute the spike primarily to depolicing. The larger share is likely driven by shifts in legitimacy, legal cynicism, economic strain, mental health, and informal social control—factors that operate through households, workplaces, street networks, and social services as much as through police behavior (McCarthy et al. 2020; Kirk & Papachristos 2011; Densley & Peterson 2024). For criminology, this implies that models of post-2020 violence should move beyond binary "police pullback vs. routine activities" narratives toward multi-level frameworks in which structural disadvantage, institutional shocks, and organizational responses jointly shape the risk of lethal conflict.

From a theoretical standpoint, our findings highlight three broader points about post-COVID, post-Floyd crime patterns. First, they highlight the importance of studying the qualities of homicide events (i.e., circumstances, victim–offender age profiles, and weapon use) rather than focusing solely on counts. The 2020 spike in Minneapolis–St. Paul is best understood not just as "more homicide" but as "more adult, argument-based homicide in disadvantaged neighborhoods," which has different implications for strain, routine-activity, and social-control theories than an across-the-board rise in domestic or felony-related killings. Second, the results demonstrate how acute legitimacy shocks interact with chronic structural inequality: delegitimizing events like Floyd's murder do not operate in a vacuum but reverberate through local ecologies in ways that magnify existing disparities. Third, the Twin Cities case shows that institutional ruptures can generate durable shifts in violence that outlast the initial moment of crisis; once patterns of conflict, retaliation, and diminished guardianship are established, they may persist even as national averages decline.

From a policy standpoint, high-profile police killings can generate collateral criminogenic harm concentrated in neighborhoods already burdened by inequality and over-policing (Phelps 2024; Vandegrift & Connor 2020). While short-term declines in proactive enforcement may weaken situational deterrence (Nagin 2013; Spelman 2017), indiscriminate re-intensification risks further erosion of state legitimacy. Evidence from randomized hot-spot and problem-oriented policing trials suggests that focused, procedurally just deployment best balances deterrence and legitimacy (Sherman & Weisburd 1995; Braga et al. 2014, 2019; Liberatore et al. 2022). Complementary community-based crisis responses—shown to reduce violence elsewhere—could be expanded to mitigate conditions that invite both police and civilian lethal encounters (Dee & Pyne 2022). More broadly, preventing future surges in homicide will require integrating structural prevention, such as addressing concentrated disadvantage and firearm access, with institutional reform to reduce excessive use of force, improve accountability, and safeguard legitimacy. In the aftermath of police violence, strategies that sustain guardianship and rebuild community trust are likely to help buffer cities from the cascading effects documented here.

Limitations

This study centers on the epicenter of the 2020 “Minneapolis reckoning” (Phelps 2024) within the context of a 30 percent national homicide increase (Gramlich 2021; Council on Crime and Justice 2023). Although similar patterns may appear in other metropolitan areas that experienced high-profile police killings, generalizations beyond Hennepin and Ramsey Counties should be approached cautiously. The interrupted-time-series design assumes that no unobserved time-varying factors coincide precisely with the treatment; while inclusion of COVID-19 caseloads, mobility, policy timing, weather, and temporal autoregressive terms support the tenability of this assumption, residual confounding cannot be ruled out. Time-varying heterogeneity in the movement of the pandemic and related behavioral responses not captured by our time-varying covariates (e.g., changes in routine activities not captured by mobility and policy controls; alterations in informal social controls temporally orthogonal to these measures, between- tract heterogeneity in pandemic response in our panel models) could confound our estimates. Therefore, the validity of our estimates of the effect of Mr. Floyd’s murder rests upon these statistical adjustments, which may not completely capture the significant homicide-related impacts of the pandemic. Further, it is beyond the scope of our current data and analysis to separate the effect of the murder from the broader context of the pandemic. Given we do not have any counterfactual observation of the murder outside of the time of a global pandemic, we are not able to speak with certainty to the effect of the murder in non-pandemic contexts (i.e., a potential moderating relationship).

Further, our causal-mediation test evaluates only one mechanism—police stops—within Minneapolis (Imai et al.2010; Tingley et al. 2014). Relationships in suburban or adjacent tracts may differ, and we cannot directly measure changes in informal social control, legal cynicism, or mental health that likely co-evolved with policing and homicide. Finally, as national data show an increasing proportion of “unknown” circumstances in homicide records (Council on Crime and Justice 2023), even hand-coded classifications remain subject to measurement uncertainty.

Overall, these caveats suggest that our estimates should be viewed as upper-bound effects in a strong-case city rather than precise forecasts for all jurisdictions. Nonetheless, the patterns we document, namely an enduring, compositionally distinctive, and spatially concentrated spike in homicide following Floyd’s murder, partially mediated by depolicing, offer a clear signal about how acute institutional ruptures can interact with chronic inequality to reshape lethal violence. For criminologists, the Twin Cities experience demonstrates the value of integrating event-level data, neighborhood context, and institutional behavior to understand how periods of crisis alter both the level and character of crime, and it points toward the development of dynamic, multi-level models capable of capturing similar patterns in other places and times.

Appendix 1

Spatiotemporal Patterns in Homicide

Figure 6 spatiotemporally visualizes the weekly homicide rates per 100,000 in each Census Tract in Hennepin and Ramsey counties. In the pre-murder time period the range of homicide rates across tracts spans from 0 to 2.77 per 100,000, with a mean of 0.122 and a standard deviation of 0.32. In contrast the range in the immediate three months post-treatment extends to 0 to 17.6, with a mean of 0.43 and a standard deviation of 1.66. This suggests that not only did the mean levels of homicide increase in Hennepin and Ramsey counties in relation to the police murder (as shown in Fig. 3), but also that the spatial variation in between-tract homicide rates increased as well. In the immediate post-treatment period, the areas of North Minneapolis, Downtown Minneapolis, the South Minneapolis neighborhoods of

Table 1 ITS models of the overall homicide rate, Hennepin/Ramsey counties 2018–2022

	Weekly Homicides/100,000			
	Base	Weather	Policy	Overall
	(1)	(2)	(3)	(4)
Time	0.001 (0.0004)	0.001* (0.0004)	0.001** (0.0004)	0.001** (0.0004)
Post-Murder	0.081* (0.038)	0.080* (0.037)	0.281*** (0.076)	0.281*** (0.076)
T Post-Murder	-0.001 (0.0005)	-0.001 (0.0005)	-0.003*** (0.001)	-0.003*** (0.001)
COVID—State of Emerg		0.001 (0.0005)	0.0003 (0.001)	0.0003 (0.001)
COVID—Stay at Home		0.011 (0.025)	0.006 (0.025)	0.006 (0.025)
COVID—School Closure		0.166 (0.091)	0.176 (0.091)	0.176 (0.091)
COVID—Business Closure			-0.088 (0.047)	-0.088 (0.047)
Mobility Pct. Change (Google)			0.150 (0.082)	0.150 (0.082)
COVID Cases			-0.066 (0.046)	-0.066 (0.046)
COVID Deaths			0.022 (0.062)	0.022 (0.062)
Mean Max. Temp			0.002 (0.002)	0.002 (0.002)
Snow (in.)			0.00000 (0.00000)	0.00000 (0.00000)
Precip. (in.)			-0.001 (0.001)	-0.001 (0.001)
AR(1)—Overall Hom	0.124* (0.061)	0.101 (0.061)	0.046 (0.063)	0.046 (0.063)
AR(2)—Overall Hom	-0.008 (0.062)	-0.038 (0.062)	-0.087 (0.063)	-0.087 (0.063)
AR(3)—Overall Hom	0.250*** (0.061)	0.232*** (0.061)	0.178** (0.063)	0.178** (0.063)
Constant	0.030 (0.028)	-0.036 (0.041)	-0.008 (0.046)	-0.008 (0.046)
ADF	-5.0**	-5.0**	-5.9**	-5.9**
KPSS	0.05	0.06	0.03	0.03
Observations	258	258	258	258
R^2	0.326	0.353	0.382	0.382
Residual Std. Error	0.143 (df=251)	0.141 (df=248)	0.140 (df=241)	0.140 (df=241)

$^{*}p\ ^{**}p\ ^{***}p < 0.001$

Table 2 ITS models of homicide rates by circumstance, Hennepin/Ramsey counties 2018–2022

	Weekly Homicides/100,000				
	Overall	Argument	Fel. Crime	Domestic	Other
	(1)	(2)	(3)	(4)	(5)
Time	0.001** (0.0004)	0.0004 (0.0002)	0.001 (0.0003)	0.0001 (0.0001)	0.00000 (0.00001)
Post-Murder	0.281*** (0.076)	0.093* (0.042)	0.095 (0.053)	-0.009 (0.019)	0.002 (0.002)
T Post-Murder	-0.003*** (0.001)	-0.001 (0.0005)	-0.001 (0.001)	0.0001 (0.0002)	-0.00002 (0.00002)
Mean Max. Temp	0.0003 (0.001)	0.0001 (0.0004)	-0.0003 (0.0004)	-0.0001 (0.0002)	-0.00000 (0.00001)
Snow (in.)	0.006 (0.025)	-0.002 (0.015)	0.016 (0.019)	0.006 (0.007)	-0.0002 (0.001)
Precip. (in.)	0.176 (0.091)	0.089 (0.054)	0.193** (0.067)	0.003 (0.025)	0.001 (0.002)
COVID—State of Emerg	-0.088 (0.047)	0.043 (0.026)	-0.005 (0.034)	-0.003 (0.013)	-0.002 (0.001)
COVID—Stay at Home	0.150 (0.082)	0.043 (0.048)	0.011 (0.060)	-0.002 (0.022)	0.002 (0.002)
COVID—School Closure	-0.066 (0.046)	-0.104*** (0.028)	-0.077* (0.035)	0.023 (0.013)	-0.001 (0.001)
COVID—Business Closure	0.022 (0.062)	0.039 (0.036)	0.017 (0.046)	-0.017 (0.017)	0.0003 (0.002)
Mobility Pct. Change (Google)	0.002 (0.002)	0.0002 (0.001)	0.001 (0.001)	0.001 (0.0004)	-0.00004 (0.00004)
COVID Cases	0.00000 (0.00000)	-0.00000 (0.00000)	-0.00000 (0.00000)	0.00000 (0.00000)	0.00000 (0.00000)
COVID Deaths	-0.001 (0.001)	-0.0004 (0.0005)	0.0001 (0.001)	0.00004 (0.0002)	-0.00001 (0.00002)
AR(1)—Overall Hom	0.046 (0.063)				
AR(2)—Overall Hom	-0.087 (0.063)				
AR(3)—Overall Hom	0.178** (0.063)				
AR(1)—Argument		-0.083 (0.063)			
AR(2)—Argument		-0.145* (0.067)			
AR(3)—Argument		0.114 (0.065)			
AR(1)—Fel. Crime			-0.067 (0.064)		
AR(2)—Fel. Crime			-0.050 (0.065)		
AR(3)—Fel. Crime			0.151* (0.065)		
Constant	-0.008 (0.046)	0.008 (0.027)	-0.002 (0.034)	0.015 (0.012)	0.0001 (0.001)
ADF	-5.0**	-7.3**	-5.9**	-5.9**	-6.54**
KPSS	0.05	0.03	0.03	0.05	0.02
Observations	258	258	258	261	261
R^2	0.382	0.203	0.181	0.069	0.024
Residual Std. Error	0.140 (df=241)	0.082 (df=241)	0.104 (df=241)	0.039 (df=247)	0.003 (df=247)

$^{*}p^{**}p^{***}p<0.001$

Table 3 ITS models of homicide rates by age and role, Hennepin/Ramsey counties 2018–2022

	Weekly Homicides/100,000				
	Overall	Youth Victim	Adult Victim	Youth Perp	Adult Perp
	(1)	(2)	(3)	(4)	(5)
Time	0.001** (0.0004)	0.0004 (0.001)	0.001* (0.001)	0.0002 (0.001)	0.001 (0.0004)
Post-Murder	0.281*** (0.076)	0.095 (0.123)	0.340*** (0.090)	0.099 (0.107)	0.225** (0.077)
T Post-Murder	-0.003*** (0.001)	-0.0004 (0.001)	-0.004*** (0.001)	0.0001 (0.001)	-0.002** (0.001)
Mean Max. Temp	0.0003 (0.001)	0.001 (0.001)	-0.00005 (0.001)	-0.001 (0.001)	0.00003 (0.001)
Snow (in.)	0.006 (0.025)	0.070 (0.045)	-0.013 (0.030)	0.026 (0.039)	-0.002 (0.026)
Precip. (in.)	0.176 (0.091)	0.026 (0.163)	0.217* (0.108)	0.137 (0.141)	0.142 (0.093)
COVID—State of Emerg	-0.088 (0.047)	0.006 (0.082)	-0.109 (0.056)	0.016 (0.071)	-0.061 (0.048)
COVID—Stay at Home	0.150 (0.082)	0.036 (0.143)	0.187 (0.096)	0.082 (0.124)	0.111 (0.084)
COVID—School Closure	-0.066 (0.046)	-0.069 (0.083)	-0.073 (0.055)	-0.098 (0.072)	-0.091 (0.048)
COVID—Business Closure	0.022 (0.062)	-0.115 (0.111)	0.064 (0.073)	0.040 (0.096)	0.025 (0.063)
Mobility Pct. Change (Google)	0.002 (0.002)	-0.001 (0.003)	0.003 (0.002)	0.001 (0.003)	0.001 (0.002)
COVID Cases	0.00000 (0.00000)	-0.00000 (0.00001)	0.00000 (0.00000)	0.00001 (0.00001)	-0.00000 (0.00000)
COVID Deaths	-0.001 (0.001)	0.002 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.0001 (0.001)
AR(1)—Overall Hom	0.046 (0.063)				
AR(2)—Overall Hom	-0.087 (0.063)				
AR(3)—Overall Hom	0.178** (0.063)				
AR(1)—Youth Victim		-0.162* (0.063)			
AR(1)—Adult Victim			0.031 (0.062)		
AR(2)—Adult Victim			-0.063 (0.064)		
AR(3)—Adult Victim			0.205** (0.063)		
AR(1)—Adult Perp					0.030 (0.061)
AR(2)—Adult Perp					-0.043 (0.062)
AR(3)—Adult Perp					0.294*** (0.062)
Constant	-0.008 (0.046)	-0.042 (0.080)	0.009 (0.055)	0.068 (0.067)	0.006 (0.047)

Table 3 (continued)

	Weekly Homicides/100,000				
	Overall	Youth Victim	Adult Victim	Youth Perp	Adult Perp
	(1)	(2)	(3)	(4)	(5)
ADF	-5.0**	-6.5**	-5.6**	-6.6**	-6.04**
KPSS	0.05	0.02	0.03	0.02	0.03
Observations	258	260	258	261	258
R^2	0.382	0.096	0.369	0.108	0.337
Residual Std. Error	0.140 (df=241)	0.251 (df=245)	0.166 (df=241)	0.219 (df=247)	0.143 (df=241)

$^{*}p^{**}p^{***}p<0.001$

Table 4 ITS models of homicide rates by weapon, Hennepin/Ramsey counties 2018–2022

	Weekly Homicides/100,000		
	Overall	Firearm	Other
	(1)	(2)	(3)
Time	0.001** (0.0004)	0.001 (0.0004)	0.0003 (0.0002)
Post-Murder	0.281*** (0.076)	0.137 (0.073)	0.150*** (0.031)
T Post-Murder	-0.003*** (0.001)	-0.001 (0.001)	-0.001*** (0.0003)
Mean Max. Temp	0.0003 (0.001)	0.0002 (0.001)	0.0001 (0.0003)
Snow (in.)	0.006 (0.025)	0.003 (0.025)	0.001 (0.011)
Precip. (in.)	0.176 (0.091)	0.185* (0.087)	0.009 (0.039)
COVID—State of Emerg	-0.088 (0.047)	-0.012 (0.046)	-0.071*** (0.020)
COVID—Stay at Home	0.150 (0.082)	0.044 (0.078)	0.113** (0.034)
COVID—School Closure	-0.066 (0.046)	-0.029 (0.044)	-0.056** (0.020)
COVID—Business Closure	0.022 (0.062)	0.042 (0.060)	-0.005 (0.027)
Mobility Pct. Change (Google)	0.002 (0.002)	0.002 (0.002)	-0.0003 (0.001)
COVID Cases	0.00000 (0.00000)	0.00000 (0.00000)	-0.00000 (0.00000)
COVID Deaths	-0.001 (0.001)	-0.001 (0.001)	-0.00003 (0.0003)
AR(1)—Overall Hom	0.046 (0.063)		
AR(2)—Overall Hom	-0.087 (0.063)		
AR(3)—Overall Hom	0.178** (0.063)		
AR(1)—Firearm		0.062 (0.063)	
AR(2)—Firearm		-0.066 (0.064)	
AR(3)—Firearm		0.096 (0.064)	
AR(4)—Firearm		-0.050 (0.064)	
AR(5)—Firearm		0.241*** (0.063)	
AR(1)—Other			-0.131* (0.062)
AR(2)—Other			-0.129* (0.062)
AR(3)—Other			-0.063 (0.062)
Constant	-0.008 (0.046)	-0.016 (0.046)	0.011 (0.020)
ADF	-5.0**	-6.0**	-5.9**
KPSS	0.05	0.02	0.03
Observations	258	256	258
R^2	0.382	0.321	0.193
Residual Std. Error	0.140 (df=241)	0.134 (df=237)	0.060 (df=241)

$^{*}p^{**}p^{***}p<0.001$

Table 5 CFA measurement model of concentrated disadvantage

	LHS	Specification	RHS	Std(Beta)	SE	*P*-Value
1	Conc. Dis	FL	Unemp. Rate	0.555	0.002	0
2	Conc. Dis	FL	Poverty Rate	0.712	0.002	0
3	Conc. Dis	FL	Female-HH Rate	0.629	0.002	0
4	Conc. Dis	FL	No HS Diploma Rate	0.898	0.002	0
5	Unemp. Rate	Cov	Poverty Rate	0.270	0.003	0

LR vs. saturated (6) = 165,410, p <.05, CFI = 1.00, SRMR =.002

Table 6 RE ITS panel models of the homicide rate, Hennepin/Ramsey counties 2018–2022

	Homicides/100,000	
	Base	Interaction
	(1)	(2)
Time	0.00000* (0.00000)	0.00000* (0.00000)
Post-Murder	0.406*** (0.099)	0.408*** (0.099)
T Post-Murder	-0.003** (0.001)	-0.003** (0.001)
Mean Max. Temp	0.001 (0.001)	0.001 (0.001)
Snow (in.)	0.008 (0.035)	0.008 (0.035)
Precip. (in.)	0.298* (0.127)	0.298* (0.127)
COVID—State of Emerg	-0.137* (0.065)	-0.137* (0.065)
COVID—Stay at Home	0.209 (0.114)	0.208 (0.114)
COVID—School Closure	-0.060 (0.065)	-0.060 (0.065)
COVID—Business Closure	0.012 (0.087)	0.011 (0.087)
Mobility Pct. Change (Google)	0.003 (0.002)	0.003 (0.002)
COVID Cases	0.00000 (0.00001)	0.00000 (0.00001)
COVID Deaths	-0.001 (0.001)	-0.001 (0.001)
Conc. Dis. (Std.)	0.149*** (0.018)	0.121*** (0.019)
AR(1)—Overall Hom	0.002 (0.003)	0.002 (0.003)
AR(2)—Overall Hom	-0.004 (0.003)	-0.004 (0.003)
AR(3)—Overall Hom	-0.006* (0.003)	-0.006* (0.003)
Conc. Dis. X Post-Murder		0.177*** (0.037)
Constant	-0.009 (0.059)	-0.007 (0.059)
Resid. Var	18.3	18.3
Tract Var	0.01	0.01
Post-Floyd Var	0.31	0.28
ADF	-52.1**	-52.0**
KPSS	0.05	0.05
Observations	122,667	122,667
Log Likelihood	-352,758.400	-352,749.700
Akaike Inf. Crit	705,560.800	705,545.400
Bayesian Inf. Crit	705,774.600	705,768.900

$^{*}p^{**}p^{***}p < 0.001$

Table 7 RE ITS Panel mediation models of the homicide rate and police stop deviations, minneapolis 2018–2022

	Police Stop Dev	Homicide Rate/100,000
	(1)	(2)
Time	-0.00001*** (0.00000)	0.00000 (0.00000)
Post-Murder	-4.421*** (0.134)	0.647* (0.304)
T Post-Murder	0.012*** (0.001)	-0.006 (0.003)
Mean Max. Temp	-0.009*** (0.001)	0.005 (0.003)
Snow (in.)	0.006 (0.049)	0.093 (0.110)
Precip. (in.)	0.145 (0.176)	0.802* (0.397)
COVID—State of Emerg	0.156 (0.090)	-0.104 (0.202)
COVID—Stay at Home	-1.873*** (0.159)	0.263 (0.358)
COVID—School Closure	0.401*** (0.090)	-0.153 (0.203)
COVID—Business Closure	-0.500*** (0.120)	-0.123 (0.271)
Mobility Pct. Change (Google)	-0.008** (0.003)	0.004 (0.007)
COVID Cases	-0.0001*** (0.00001)	0.00000 (0.00002)
COVID Deaths	0.023*** (0.002)	-0.001 (0.003)
Conc. Dis. (Std.)	0.152** (0.056)	0.284*** (0.057)
Police Stop Dev		-0.050*** (0.012)
AR(1)—Overall Hom	-0.005* (0.003)	0.009 (0.006)
AR(2)—Overall Hom	-0.007** (0.003)	-0.010 (0.006)
AR(3)—Overall Hom	-0.008** (0.003)	-0.009 (0.006)
Constant	0.683** (0.250)	-0.308 (0.203)
Resid. Var	9.03	45.96
Tract Var	6.55	0.36
ADF	-32.8**	-32.5**
KPSS	0.05	0.05
Observations	31,218	31,218
Log Likelihood	-79,020.660	-104,168.800
Akaike Inf. Crit	158,081.300	208,379.600
Bayesian Inf. Crit	158,248.300	208,554.900

$^{*}p^{**}p^{***}p<0.001$

Phillips and East Phillips, as well as a few suburban tracts showed significant increases in the homicide rate after the police murder of George Floyd. In the period following the three months post-murder, the tract that exhibited increases in the homicide rate showed reduced levels in relation to the treatment period (mean = 0.31, SD = 0.75). In sum, the maps in Appendix Fig. 6 indicate that the impact of the police murder on the homicide rate was not experienced equally across all tracts, and that the deleterious crime effects of the murder are concentrated in a subset of Census Tracts.

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Data Availability A Github Repository (https://github.com/ryanplarson/CNOH-Replication-Repository) contains complete R code and partial data to replicate the findings in this study. The hand-coded BCA data from the Violence Prevention Project is not publicly available, so only aggregated data is included in this repository. Therefore, code to build these data will not work without an access link, which can be requested at violence-prevention@hamline.edu. Replication code will also require Google and Census Bureau API keys. In addition, the Google Mobility Reports data is too large to house in the repository, but is publicly available at https://www.google.com/covid19/mobility/. Further, the pre-built time series and panel datasets are available upon request if replicators wish to skip the data build section of the script.

Declarations

Competing interest The authors declare that they have no competing interests.

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